

PROJECT: STM32-PLC
Sub-system: DA1_prj.PrjPcb
Version: 1.0.0



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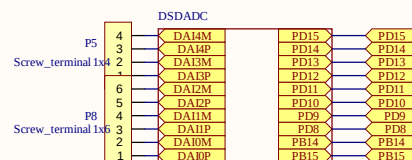
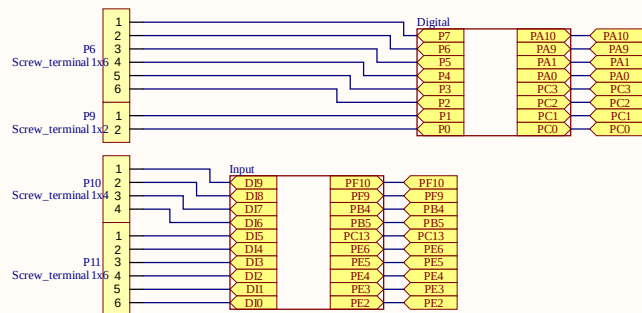
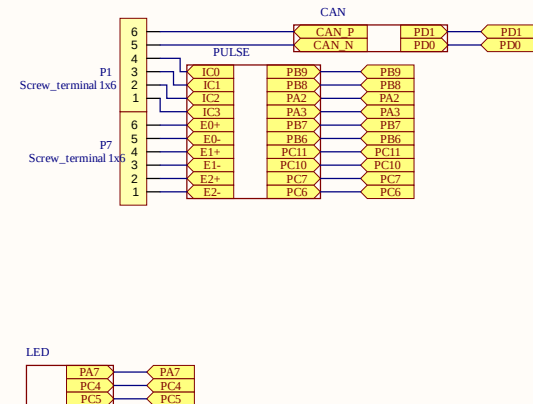
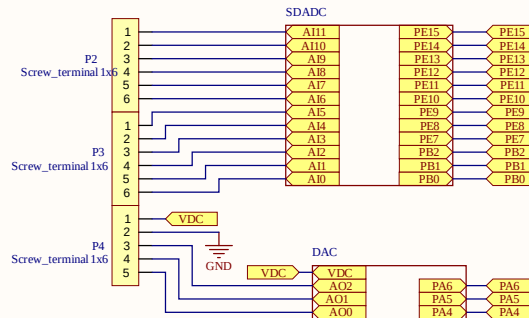
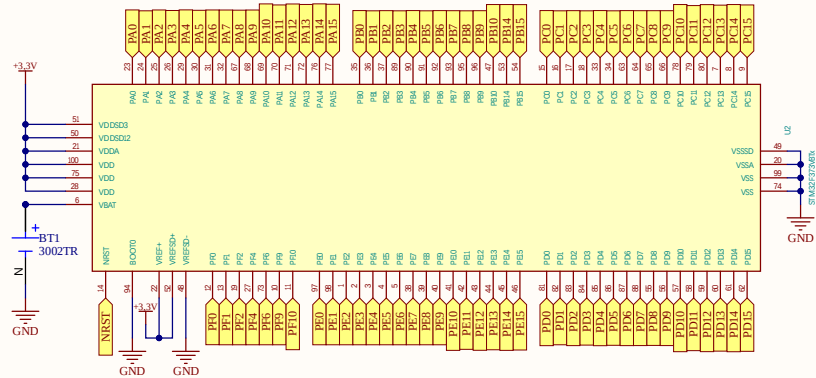
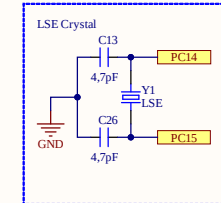
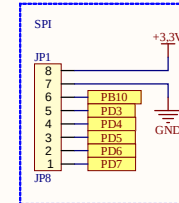
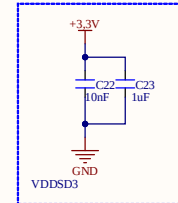
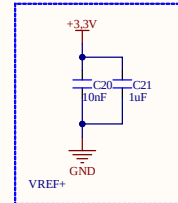
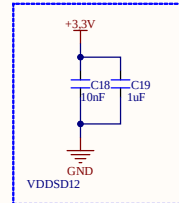
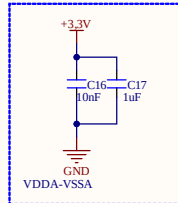
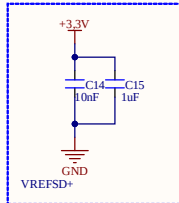
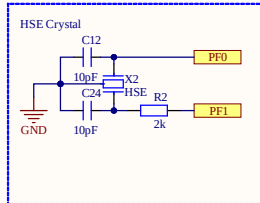
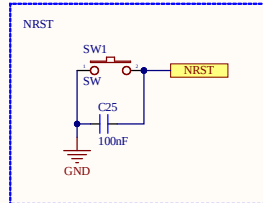
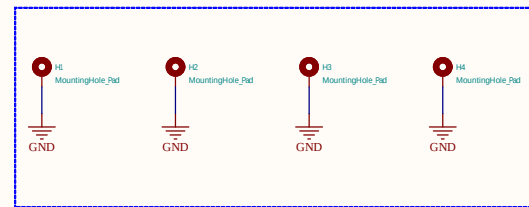
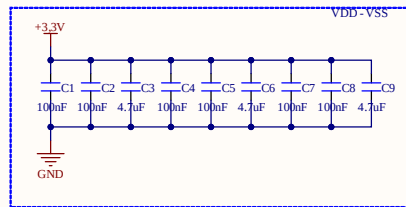
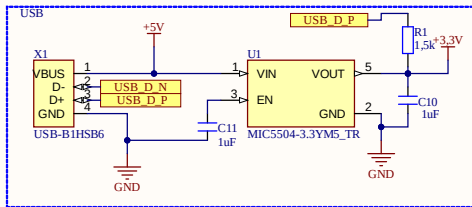
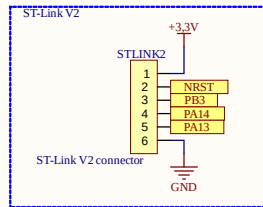
Sub-system: DA1_prj.PrjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: COVER PAGE	Sheet 1 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 1	Size: A3
Approved by: *	Revision: 1.0.0	Dates: March 14

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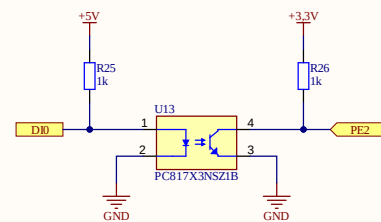
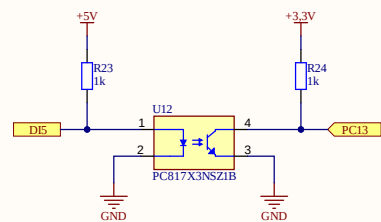
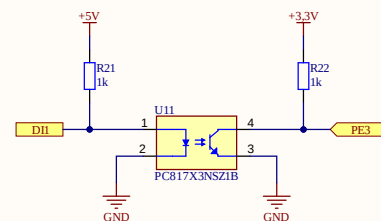
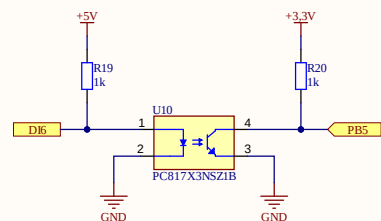
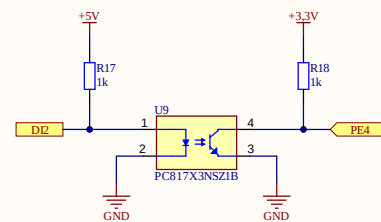
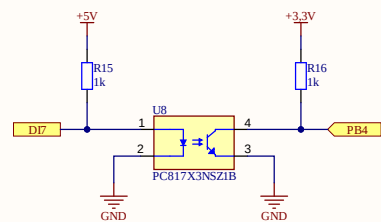
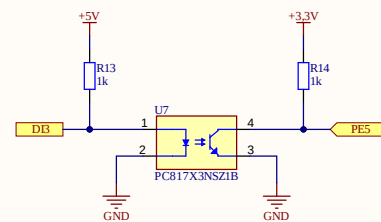
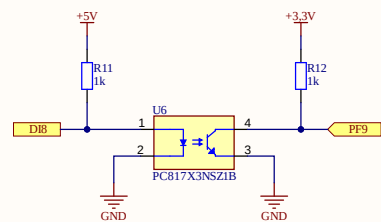
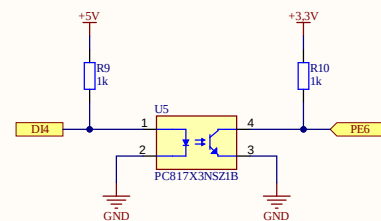
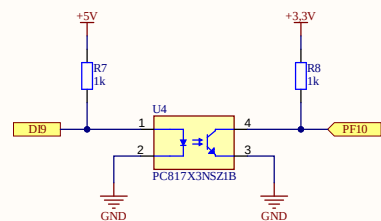
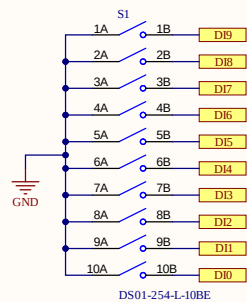
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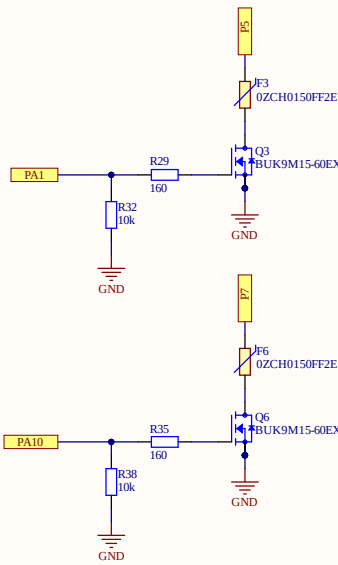
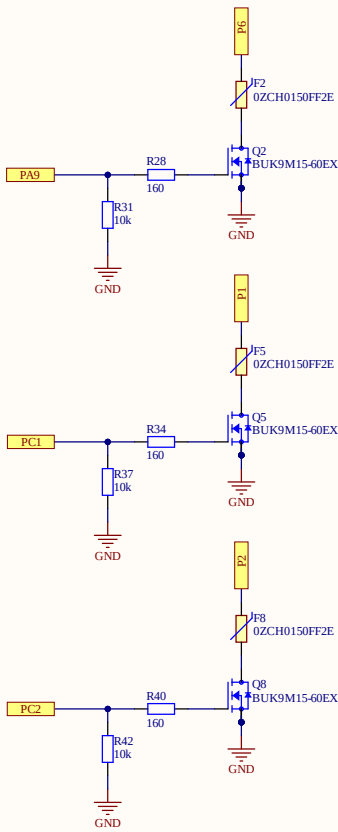
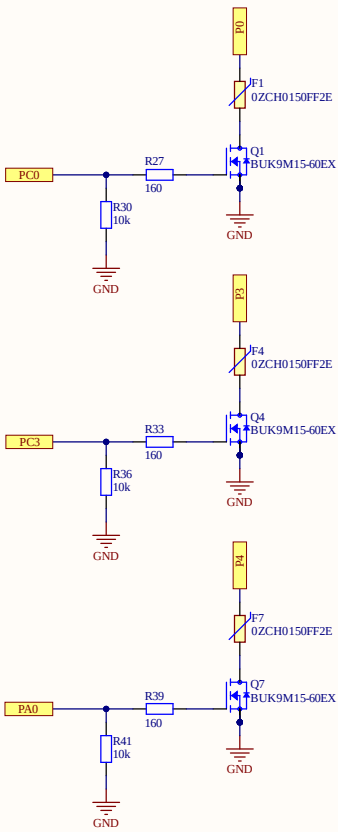


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Sub-system: DA1_pj,PjPdb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: STM32_PLC	Sheet 3 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 3	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14



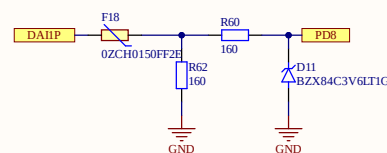
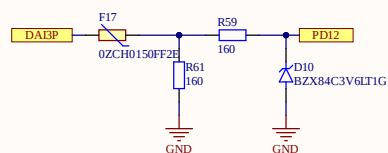
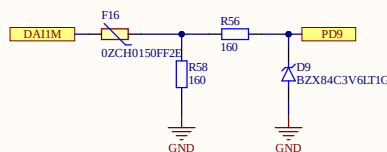
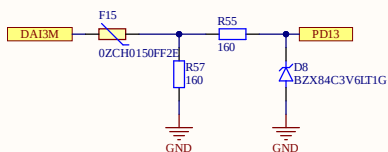
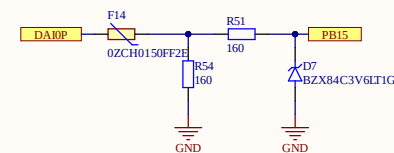
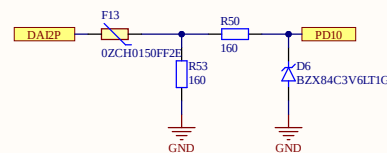
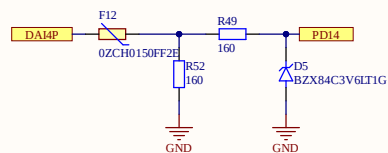
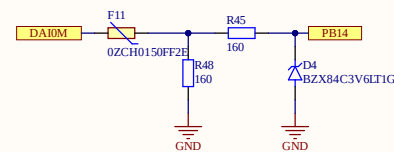
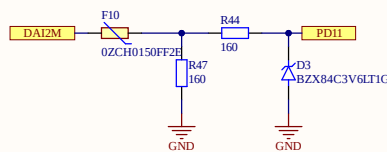
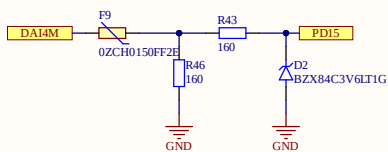
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Sub-system: DA1_prj.PrjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: INPUT	Sheet 4 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 4	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14



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Sub-system: DA1_prj,PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: DIGITAL	Sheet 5 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 5	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14

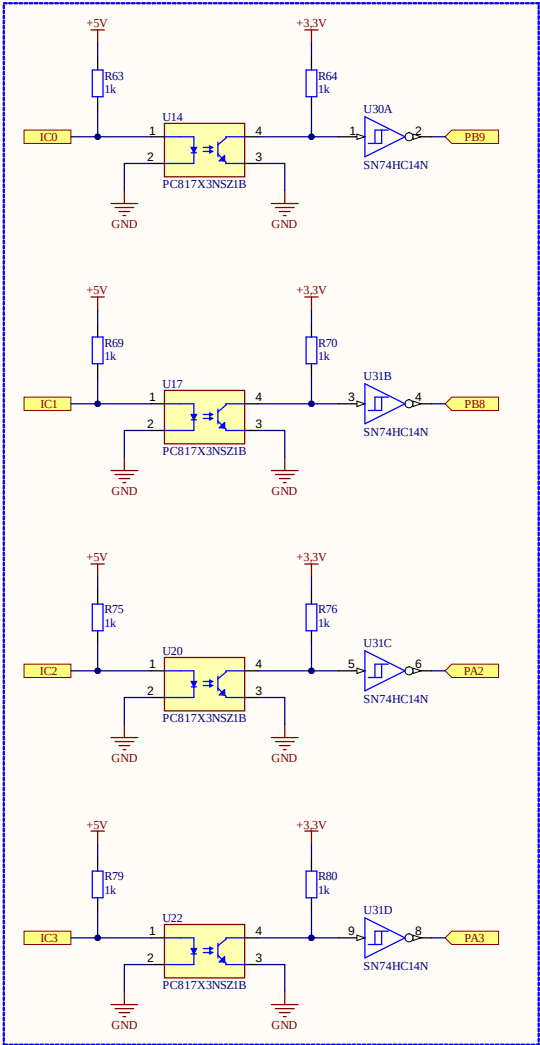
Here we convert 4-20mA to 0.64-3.2V due to the 160 ohm resistor.
Protection is available due to the polyfuse and the 160 ohm resistor for
protecting the zener diode and the zener diode protecting the analog input of the controller.



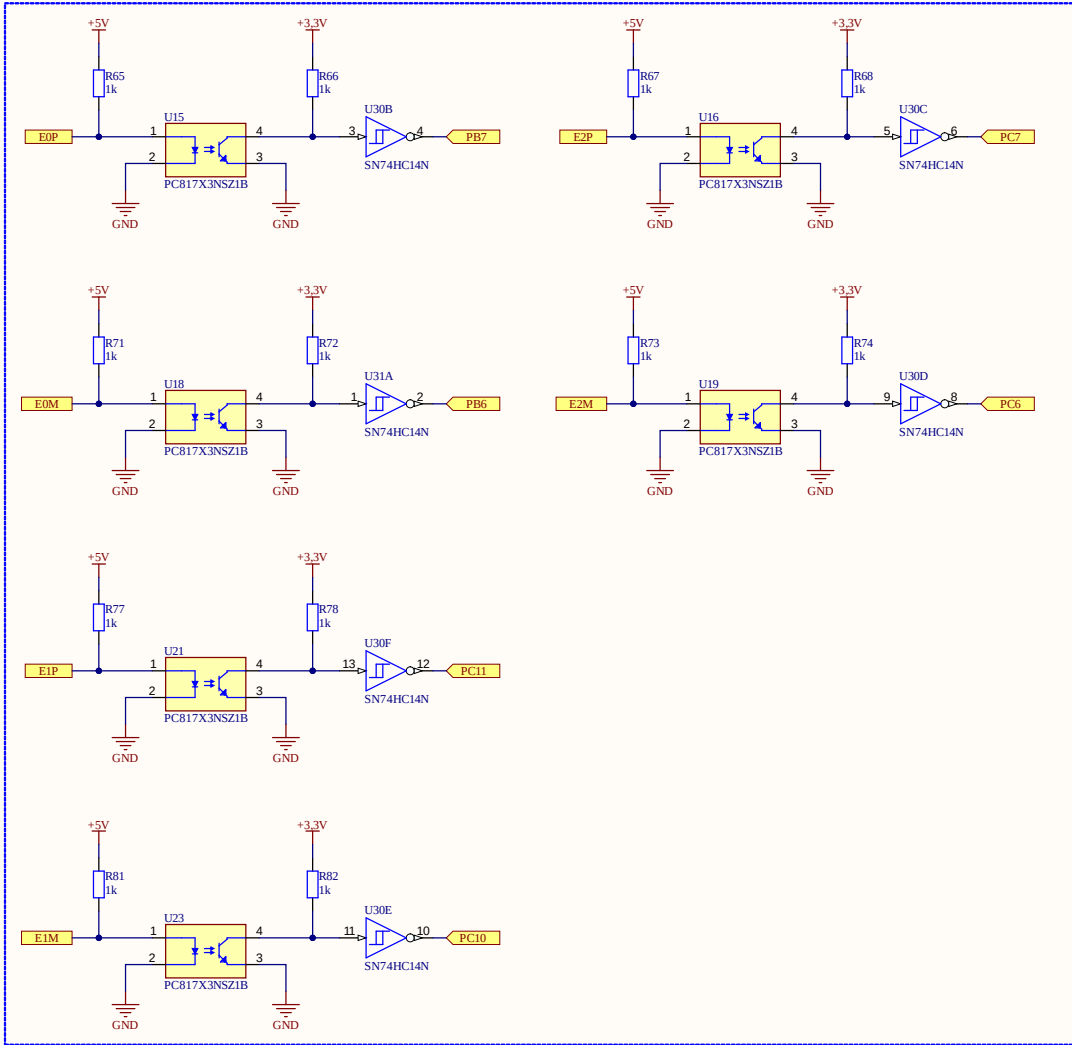
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Sub-system: DA1_prj,PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: DSDADC	Sheet 6 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 6	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14

INPUT CAPTURE

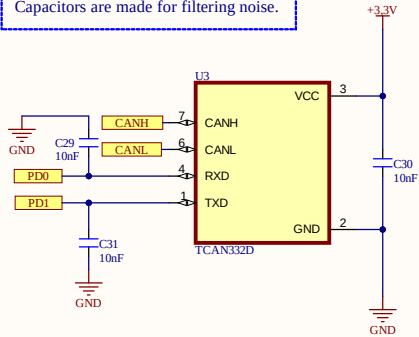


ENCODER

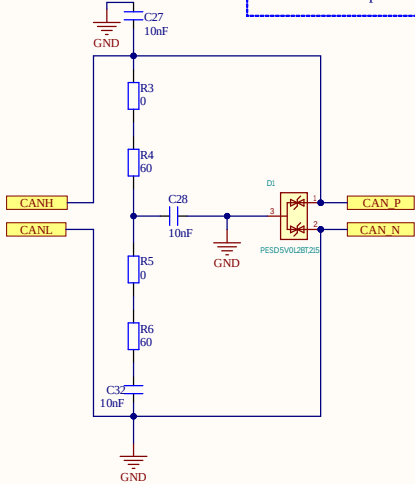


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Sub-system: DA1_prj,PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: PULSE	Sheet 7 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 7	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14

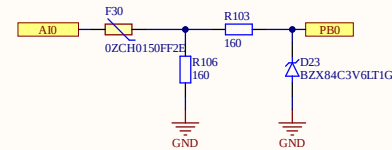
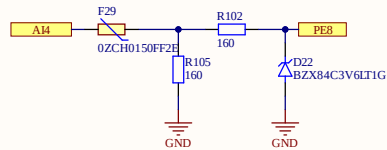
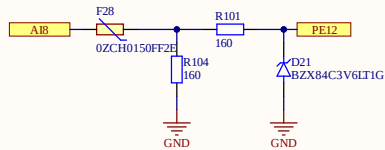
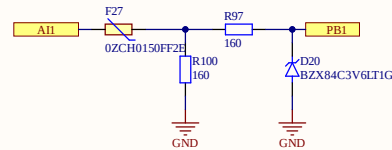
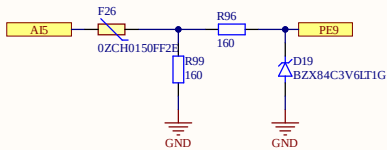
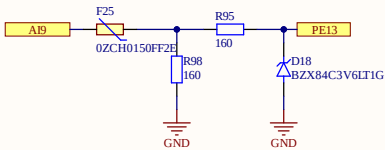
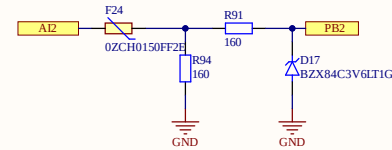
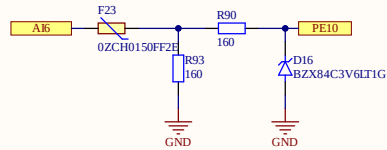
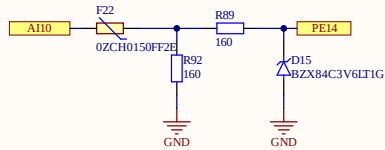
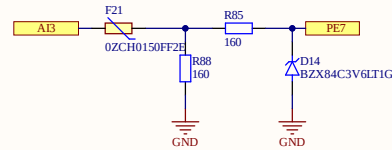
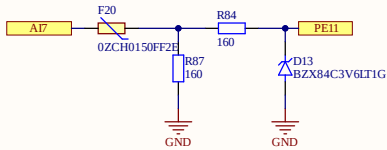
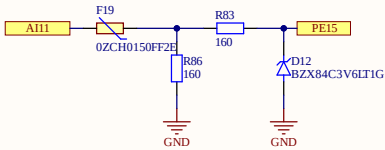
Capacitors are made for filtering noise.



CAN-bus with split terminal + two diodes for high voltage protection.

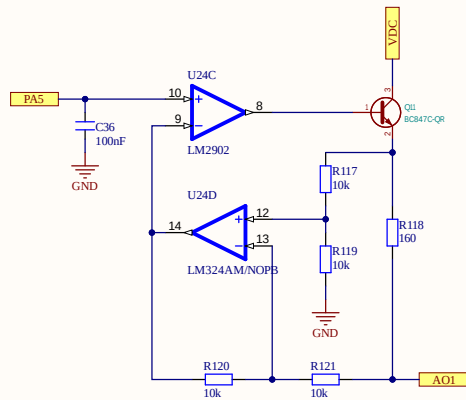
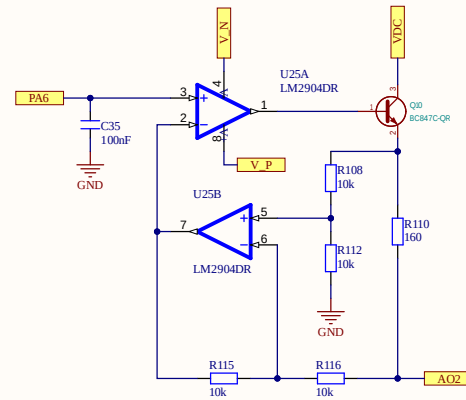
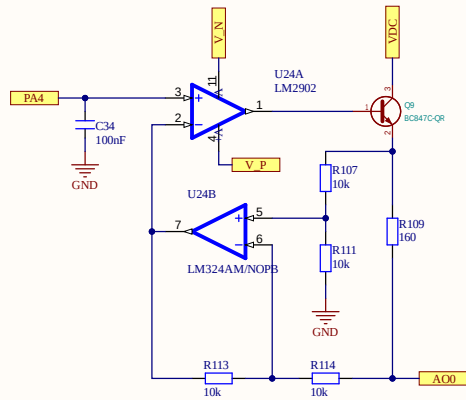
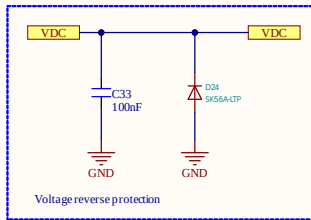
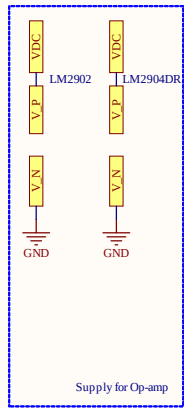


BK TP-HCM VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY		
Sub-system: DA1_prj.PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: CAN	Sheet 8 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 8	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14



Here we convert 4-20mA to 0.64-3.2V due to the 160 ohm resistor.
Protection is available due to the polyfuse and the 160 ohm resistor for
protecting the zener diode and the zener diode protecting the analog input of the controller.

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Sub-system: DA1_prj,PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: SDADC	Sheet 9 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 9	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14



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Sub-system: DA1_prj.PjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: DAC	Sheet 10 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 10	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14

A

B

C

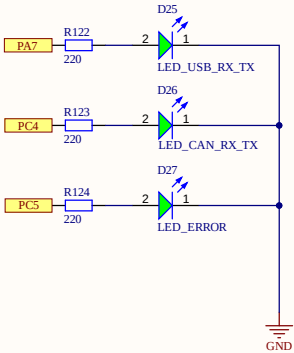
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A

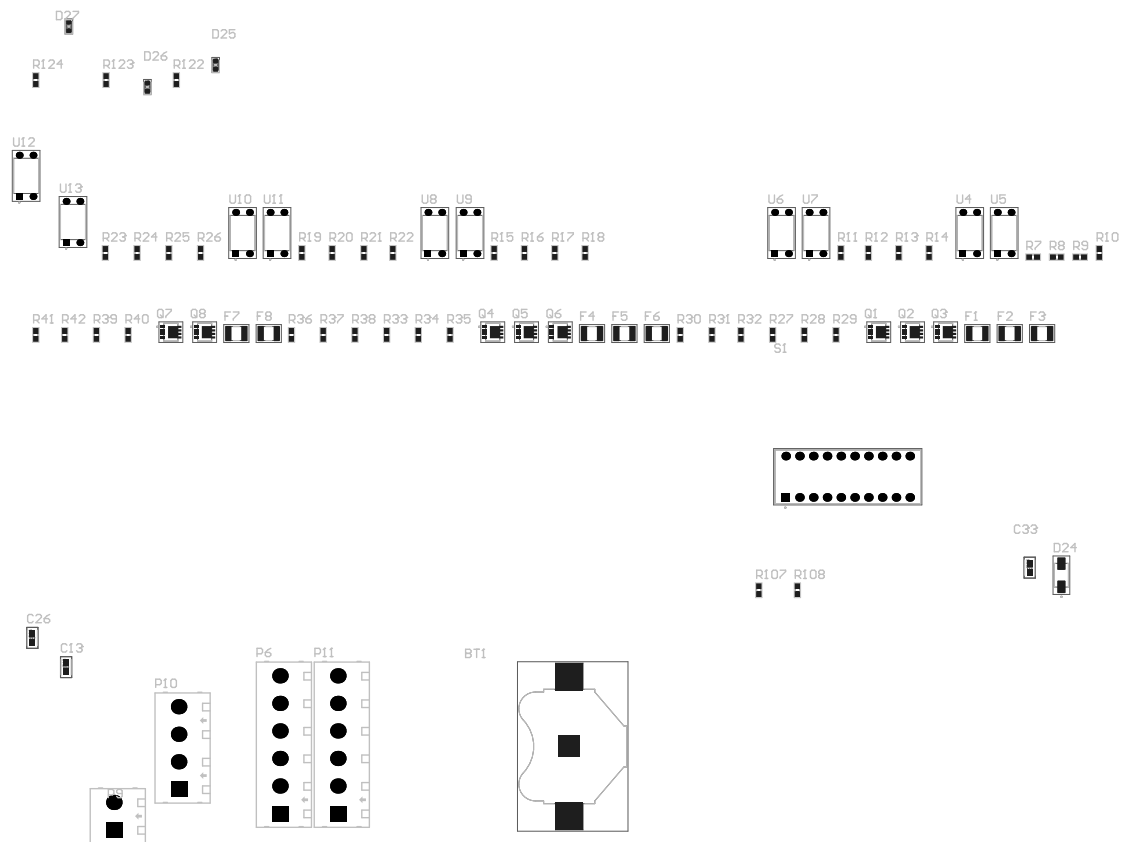
B

C

D



BK TP-HCM VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY		
Sub-system: DA1_prj.PcjPcb	Project: STM32-PLC	
Design by: Nguyễn Gia Huy	Sheet title: LED	Sheet 11 of 11
Checked by: ThS. Bùi Quốc Bảo	Doc No: 11	Size: A3
Approved by: *	Revision: V1.0	Dates: March 14



[illegible]